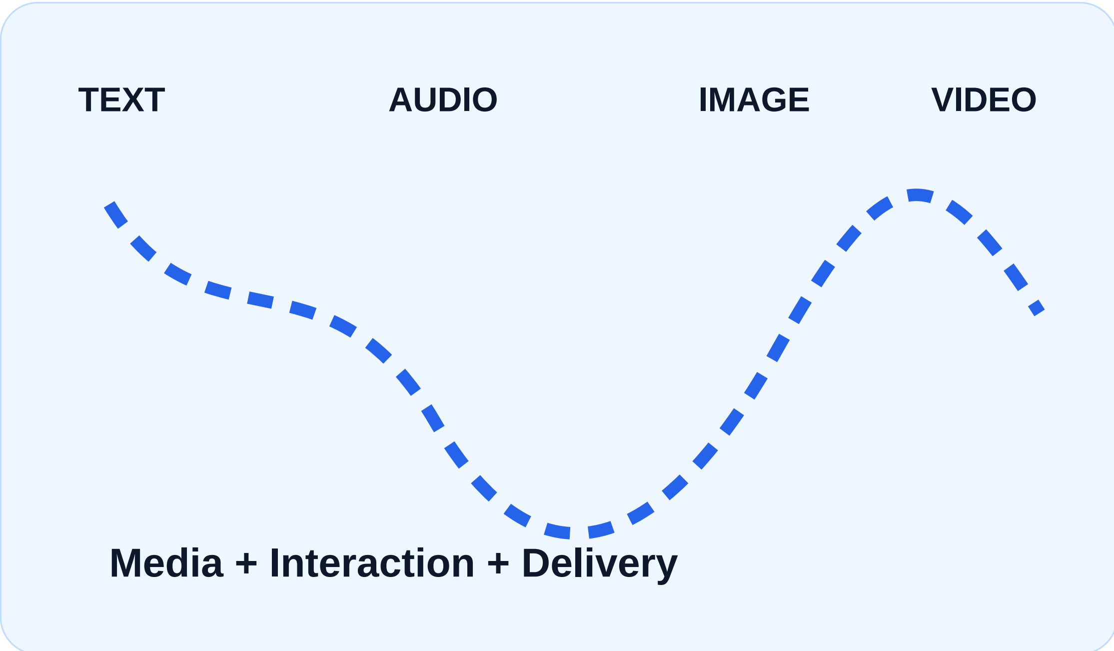
Open Elective - Electronics / ECE

Introduction to Multimedia

Corrected lesson path for **Course Code 6042D**. This handbook covers multimedia components, project stages, text, hypertext, digital audio, MIDI, compression, images, colour, video, animation and multimedia networks.

Course Code
6042D

Correct file
lessons-6042D.html



Course overview

Multimedia is not just presentation design. It is the engineering of different media objects so that they can be created, stored, compressed, transmitted, synchronized and delivered to users.

> JPEG, PNG, MIDI, MPEG □□□□□□□□ names □□□□□□□ □□□□□□□□□ □□□□. □□□ format □□□□□□□ □□□□□□□□□□□□, quality/file size/bandwidth □□□□□□□ □□□□□□ □□□□□□□□□ exam-□□□ practical-□□□ important.

CO	Focus	Practical target
CO1	Components and applications	Explain multimedia objects, applications, digital transition and project stages.
CO2	Text/audio/compression	Explain fonts, hypertext, digital audio, MIDI and compression methods.
CO3	Images	Explain colour models, palette, dithering, GIF, JPEG, PNG and TIFF.
CO4	Video and networks	Explain video compression, MPEG-4, animation, conferencing and MOD.

Module 1 - Multimedia basics and project stages

Definition

Multimedia is the integrated use of text, graphics, image, animation, audio and video under computer control. Interactive multimedia lets the user control navigation.

Components

Text, graphics, images, audio, video, animation and interactivity. Each component must be selected based on purpose, quality and delivery platform.

Digital transition

Digital media allows editing, compression, search, copying, storage, networking, indexing and interaction compared with conventional fixed media.

Project stages

Requirement → planning → storyboard → content creation → authoring → testing → delivery → maintenance.

Tools

Input devices include camera, microphone and scanner. Production tools include image editor, audio editor, video editor, animation software, presentation tools and authoring systems.

Module 2 - Text, audio and compression

Text and fonts

Text provides headings, navigation, captions, instructions and accessibility. Bitmap fonts are pixel based. Outline fonts are curve based and scalable. Unicode supports international character sets.

Hypertext

Hypertext is linked text. Hypermedia extends links to image, audio, video and animation. Node is an information unit; anchor is the clickable source; link connects nodes.

Digital audio

Digital audio is created by sampling and quantizing an analog sound wave. Important terms: frequency, bandwidth, decibel, sampling rate, bit depth and channels.

> Audio data rate = Sampling rate × Bit depth × Channels

MIDI

MIDI stores musical instructions, not actual waveform. It is small and useful for synthesized music. Digitized audio stores actual sound samples.

Compression methods

Lossless compression recovers original data exactly. Lossy compression removes less noticeable information. Methods include RLE, Huffman coding, dictionary coding, arithmetic coding, sliding window compression and silence compression.

Module 3 - Image fundamentals

Image basics

A digital image is a 2D array of pixels. Quality depends on resolution, colour depth, dynamic range, display and compression.

> Image size = Width × Height × Bits per pixel / 8

Colour

RGB is used for displays, CMYK for printing, HSV/HSL for colour selection and grayscale for intensity-only images. Palette means limited colour table. Dithering simulates unavailable colours with patterns.

Formats

Format	Best use
GIF	Simple graphics/animation
JPEG	Photographs
PNG	Diagrams/screenshots
TIFF	High-quality scanning/archive

JPEG idea

JPEG divides the image into blocks, applies transform coding, quantizes details and entropy-codes the result. It is lossy and suitable for photos, not sharp text diagrams.

Module 4 - Video, animation and networks

Video basics

Video is a sequence of frames. Important terms: frame rate, resolution, aspect ratio, bitrate, codec and container.

> Raw video rate = Width × Height × Bits per pixel × Frames per second

Compression

Video compression removes spatial redundancy within frames and temporal redundancy between frames. Motion compensation stores movement and differences instead of every full frame.

Animation

Cell animation uses frame-by-frame drawings. Computer animation uses keyframes and interpolation. Morphing transforms one image/object into another.

Networks

Multimedia networks require bandwidth, low delay, low jitter, low packet loss, buffering and synchronization. Applications include multimedia over IP, ATM, conferencing and media-on-demand.

Formula and file bank

Compression ratio = Original size / Compressed size

Saving % = (Original - Compressed) / Original × 100

Audio storage = Data rate × Duration / 8

Format logic: photo - JPEG, diagram - PNG, animation icon - GIF, archive scan - TIFF, audio edit - WAV, compressed music - MP3/AAC, music instructions - MIDI, common video - MP4.

Question bank

1. Define multimedia and explain its components.
2. Explain transition from conventional media to digital media.
3. Explain stages of a multimedia project.
4. Compare bitmap and outline fonts.

5. Define hypertext, hypermedia, node, anchor and link.
6. Explain audio digitization and data rate.
7. Compare MIDI and digitized audio.
8. Explain lossless and lossy compression with examples.
9. Explain colour models, palette and dithering.
10. Compare GIF, JPEG, PNG and TIFF.
11. Explain motion compensation in video compression.
12. Explain multimedia over IP and media-on-demand.

Model paper

Part A: Definitions - multimedia, hypertext, sampling, compression ratio, dithering, motion compensation, MOD.

Part B: Project stages, fonts, MIDI vs audio, compression methods, image formats, animation methods.

Part C: Full answers on multimedia components, audio digitization, image compression, video compression and multimedia networks.